Automation Touch Point Agreement

1.0 Touch Point Design

The automation touch point definition details the automation transaction points at each operation where data will be collected relating to part production.

1.1 Standard MES Production Station Automation Touch Points

The following table defines the standard Manufacturing Director™ Automation Touch Points.

Touch Point Names	MES Read/Write	Data Type	Description
DataReady	Read/Write	Bool	Data Ready Trigger: • Set by automation when process data is ready; typically when a part is complete at an operation • Reset by MES on start of transaction
TransInProc	Read/Write	Bool	Transaction In Process • Set by MES on start of transaction • Reset by MES on completion of transaction
PartSN	Read	String	Serial Number is set in automation and is to be collected by MES upon DataRdy.
PartStatus01	Read	Bool	Part Status is set in automation and is to be collected by MES upon DataRdy. This determines if the part/feature is a Pass (True) or Fail (False). Failed parts are not added to a container and are to be scrapped or reworked. This is not a required field and will be applicable by station.

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HardwareInterlockEnable	Read	Bool	Hardware Interlock Enable –Set in the MES Interface or in the MIP (Machine Interface Panel). This bit will Enable or Disable the MES checks executed at the automation level (i.e., quality checks, serial number processing, etc…). MES may still be able to process parts if the Hardware Interlock is disabled.
MESInterlockEnable	Read/Write	Bool	MES Interlock Enable –Set in the MES Interface. This bit will Enable or Disable the MES checks (i.e., duplicate serial number, serial number format validation, etc.) executed at the MES software level MES will not record any information if the MES Interlock is Disabled.
PartValid	Write	Bool	MES Station Validation – this point indicates that all rules and logic defined for the station have passed and the part is valid to continue through the process. MES sets this value True when validation is passed. This is required for the part to continue the process.
WatchDog	Read/Write	Bool	WatchDog touch point provides indication between automation and MES that device connections are present and ready for machine integration.
AudibleAlertEnable	Read/Write	Bool	MES Audible Alert Enable –Set at the MIP HMI or in the MES Interface. This bit will Enable or Disable the Audible Alert on the MIP in the event a part fails validation (MESStnInvalid) or a quality check.
ContainerCount	Write	Integer	Part Count in current Container
ReqContainerCount	Read	Bool	Request container count at MIP to MES
PartDisp	Write	String	Status of the part.

PartType	Write	String	Part Type being run at station
AlarmMsg	Write	String	Alarm from MES Transaction

1.2 MES Production Station – Automation Touch Point Map

The following table identifies the production stations and the touch points that apply to each station in addition to the data sources and the host stations:

Touch Point Names	5GO - Assembly Fronts	5GO - Assembly Rears	PNA - Assembly Out CH IN/EX 5,6 FP	PNA - Assembly Out 1 RH 1-4 CH IN/EX 1-4
DataReady	Yes	Yes	Yes	Yes
TransInProc	Yes	Yes	Yes	Yes
PartSN	Yes	Yes	No	No
PartStatus01	No	No	No	No
HardwareInterlockEnable	Yes	Yes	Yes	Yes
MESInterlockEnable	Yes	Yes	Yes	Yes
PartValid	Yes	Yes	Yes	Yes
WatchDog	Yes	Yes	Yes	Yes
AudibleAlertEnable	Yes	Yes	Yes	Yes
ContainerCount	Yes	Yes	Yes	Yes
ReqContainerCount	Yes	Yes	Yes	Yes
PartDisp	No	No	No	No
PartType	Yes	Yes	Yes	Yes
AlarmMsg	Yes	Yes	Yes	Yes

1.3 Machine Integration to MES

Machine Integration to the MES is accomplished with an intermediate Machine Integration Panel (MIP) between the machine automation and the MES. The MIP allows for minimal changes to be made to the machine automation hardware and logic for the MES interaction required. The machine integration diagram below details what is required for the stations in scope.

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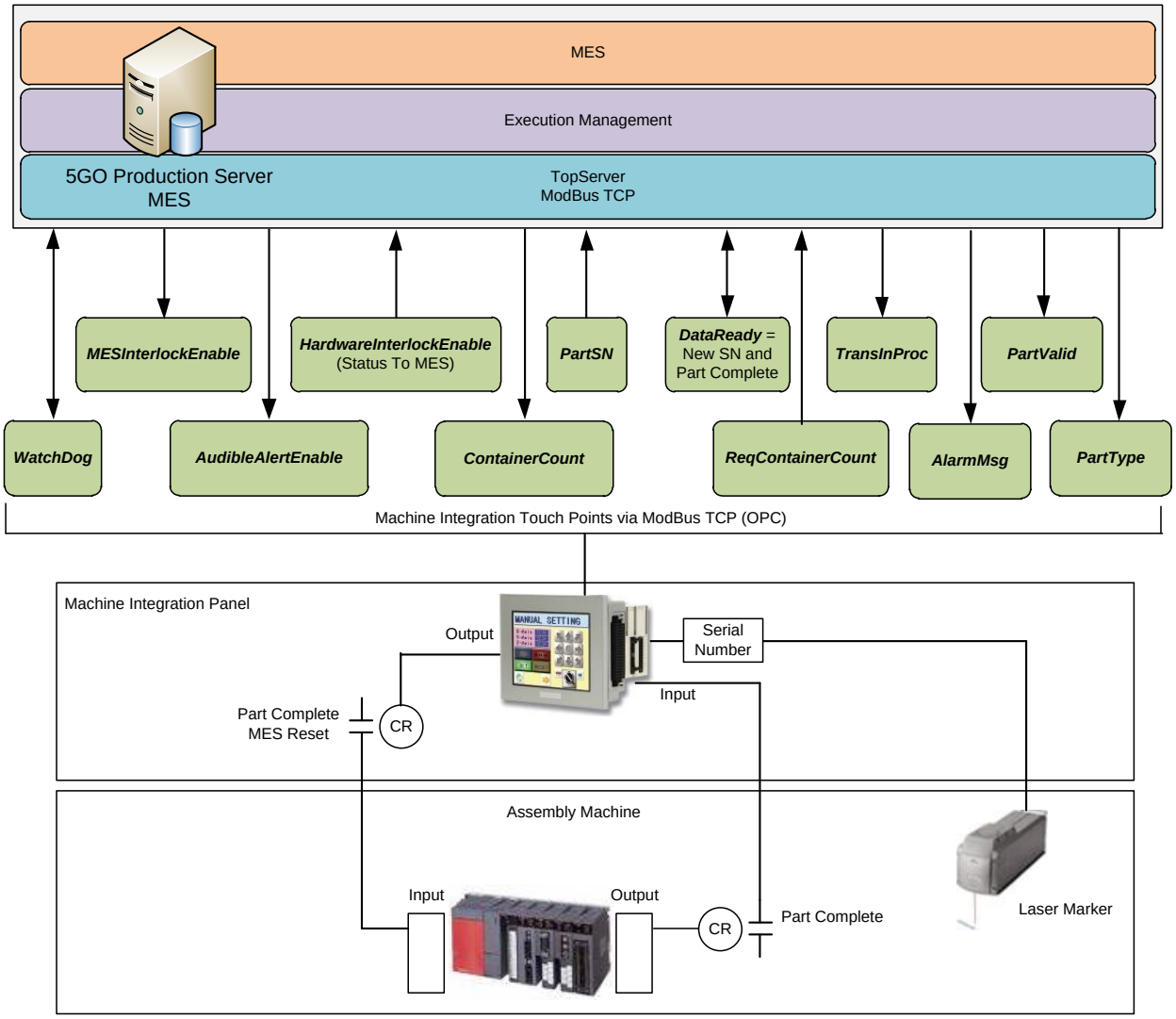
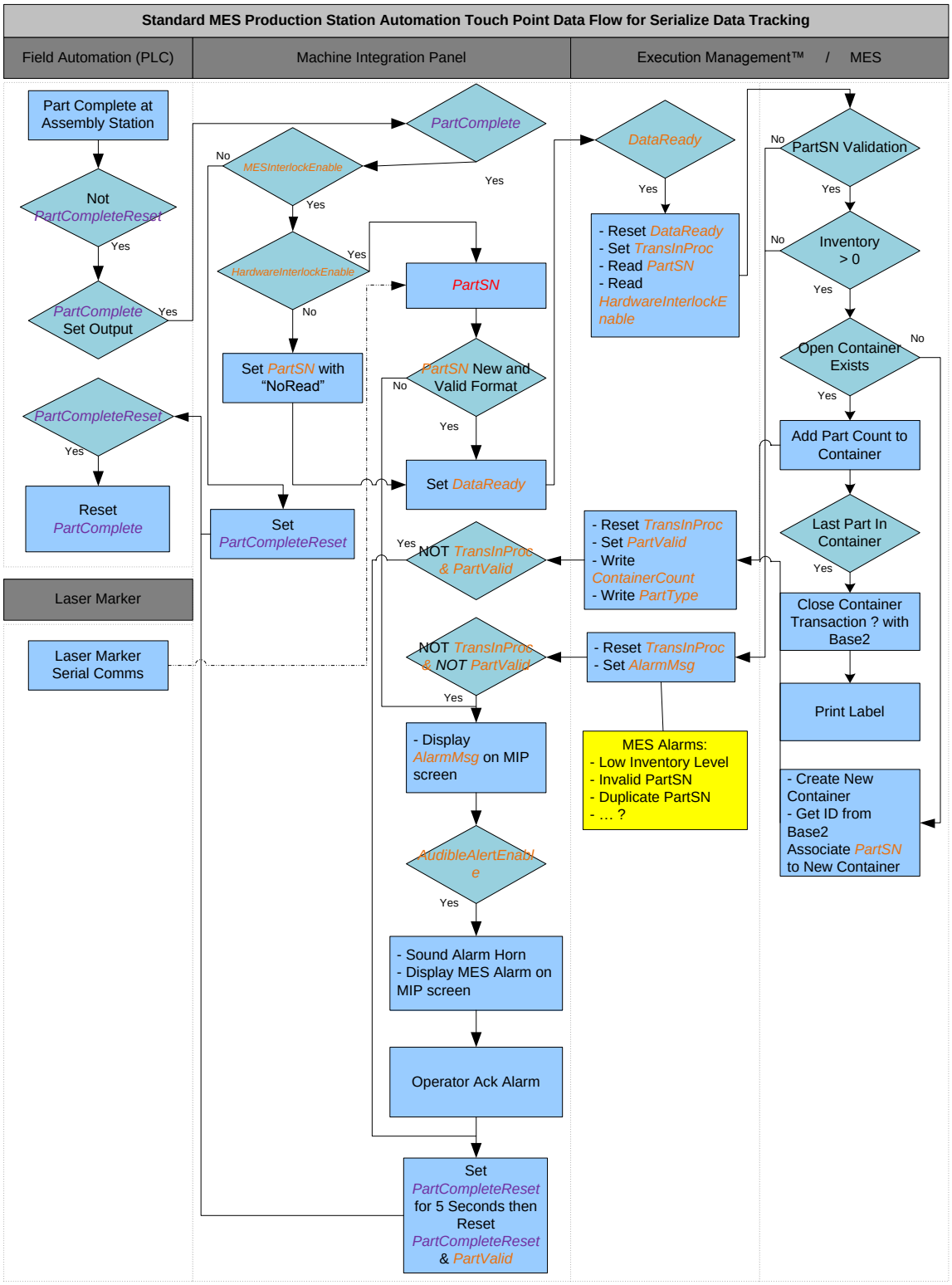

MES
Execution Management
5GO Production Server
MES
TopServer
ModBus TCP
MESInterlockEnable
HardwareInterlockEnable
(Status To MES)
PartSN
DataReady =
New SN and
Part Complete
TransInProc
PartValid
WatchDog
AudibleAlertEnable
ContainerCount
ReqContainerCount
AlarmMsg
PartType
Machine Integration Touch Points via ModBus TCP (OPC)
Machine Integration Panel
Output
Serial
Number
Input
Part Complete
MES Reset
CR
Assembly Machine
Input
Output
CR
Part Complete
Laser Marker

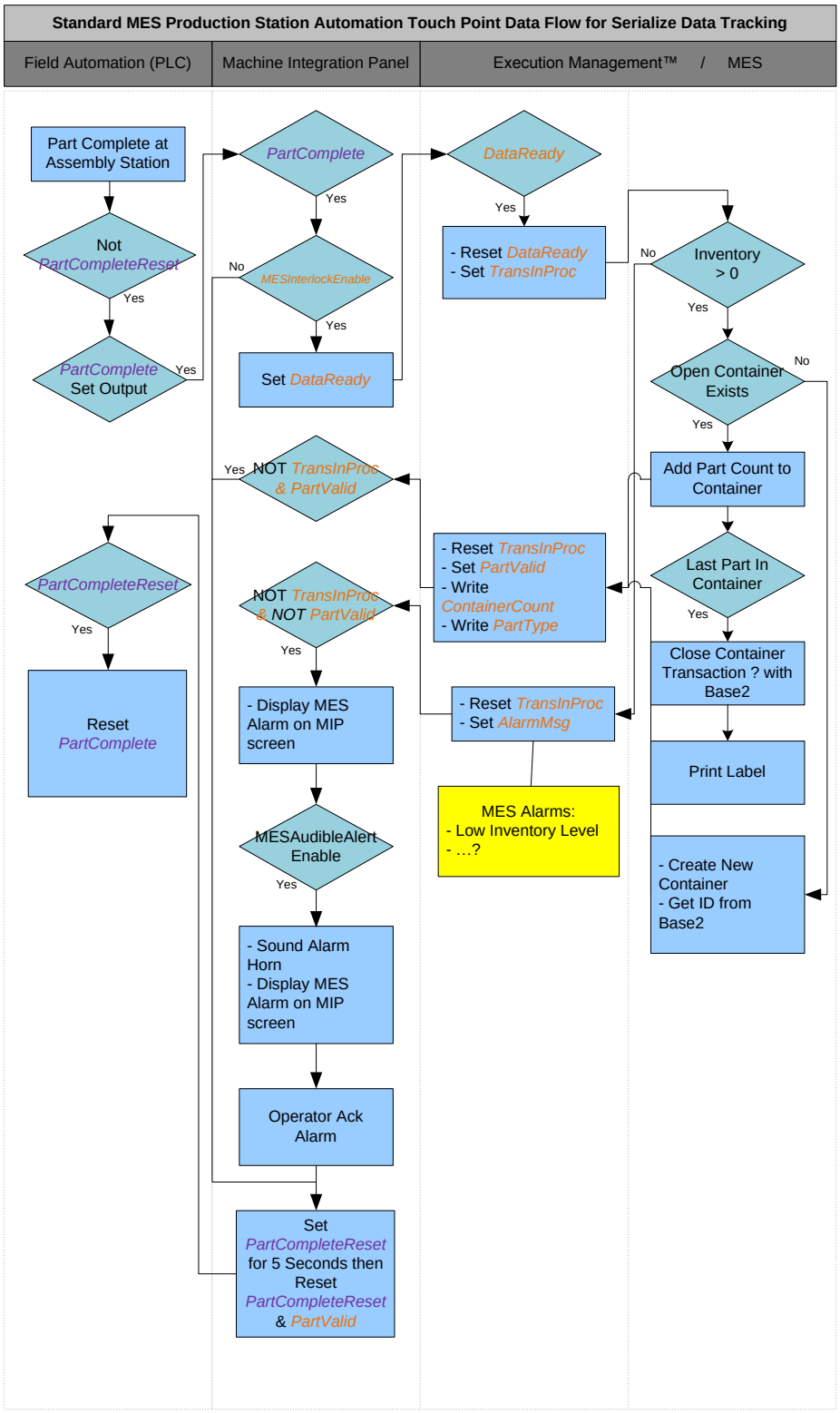
1.4 5G0 – Part Complete Automation > MES Flow Diagram

The following touch point flow diagram defines the interaction between Automation, Machine Integration, and MES for a part complete transaction with a Serial Number.



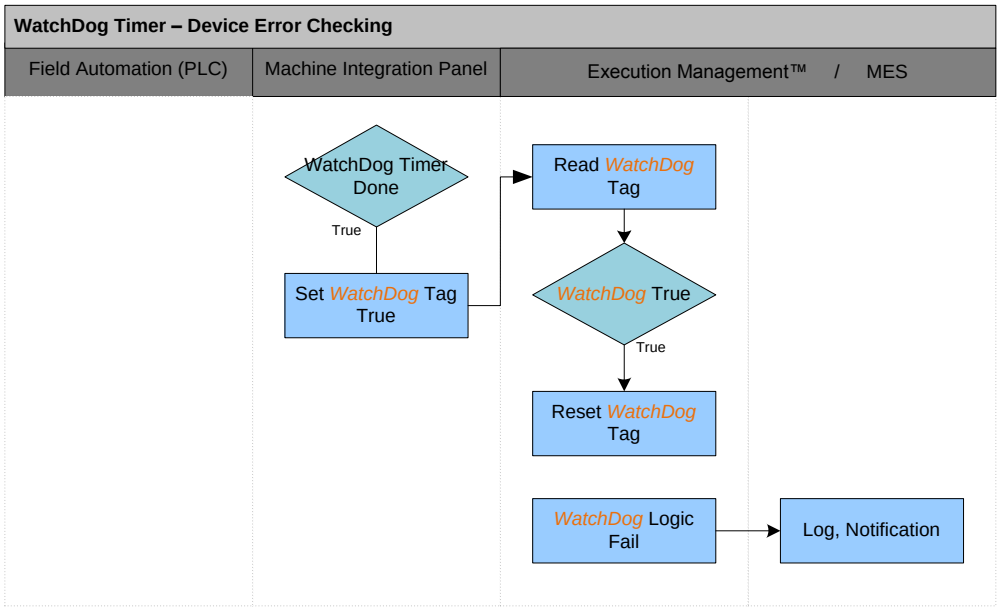
1.5 PNA – Part Complete Automation > MES Flow Diagram

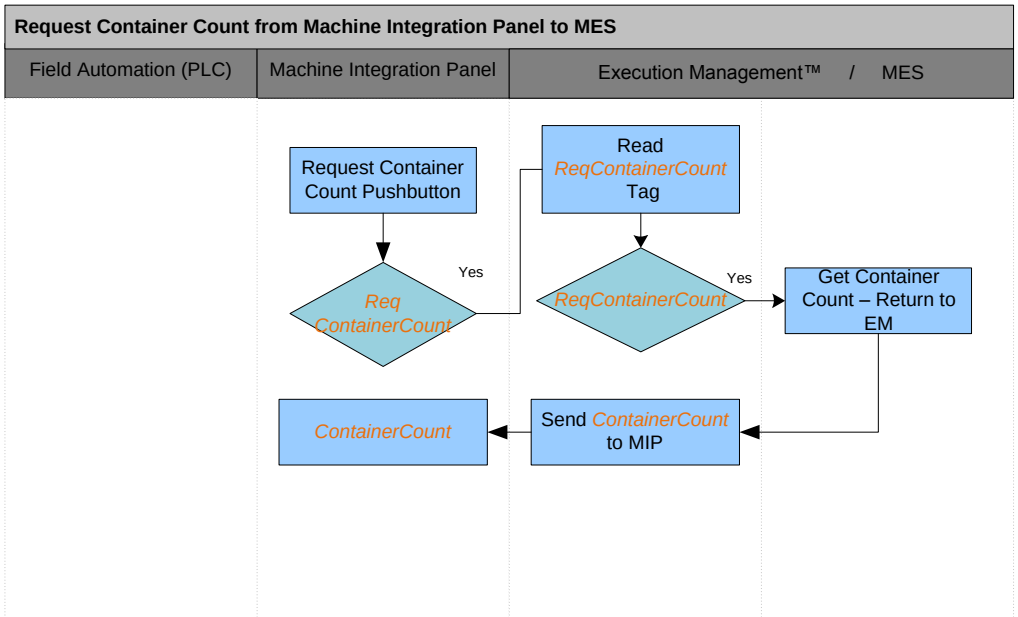
The following touch point flow diagram defines the interaction between Automation, Machine Integration, and MES for a part complete transaction based on lot tracking.



1.6 WatchDog Timer – Device Error Checking

The following touch point flow diagram defines the interaction between the machine integration panel and MES for WatchDog handshaking.



1.7 Container Count Request

The following touch point flow diagram defines the interaction between the machine integration panel and MES for the request of the container count to the machine integration panel.